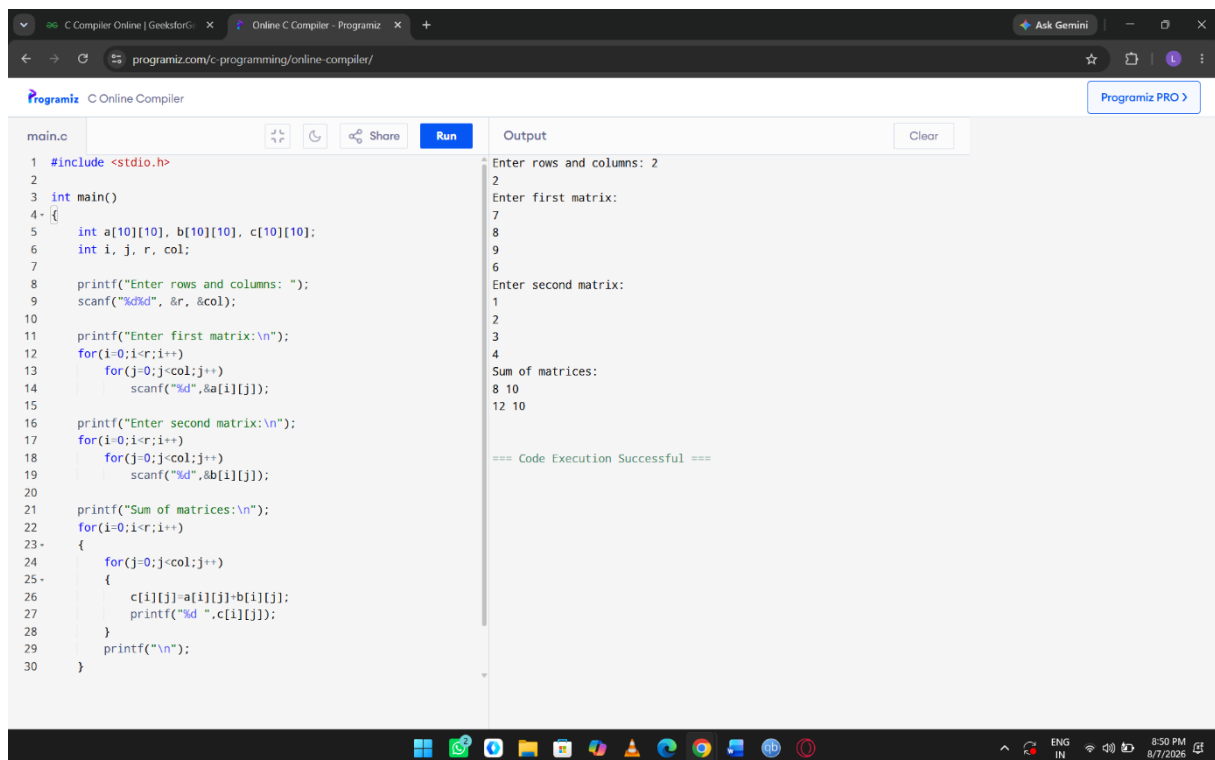


GITHUB ASSISGNNMENT-1

IT23302 DATA STRUCTURES

1.Sum of two matrices



The screenshot displays a web browser window with the URL `programiz.com/c-programming/online-compiler/`. The browser tabs include "C Compiler Online | GeeksforGeeks" and "Online C Compiler - Programiz". The page title is "Programiz C Online Compiler". The code editor shows a C program for summing two matrices. The output window shows the execution results, including the input matrices and the resulting sum matrix.

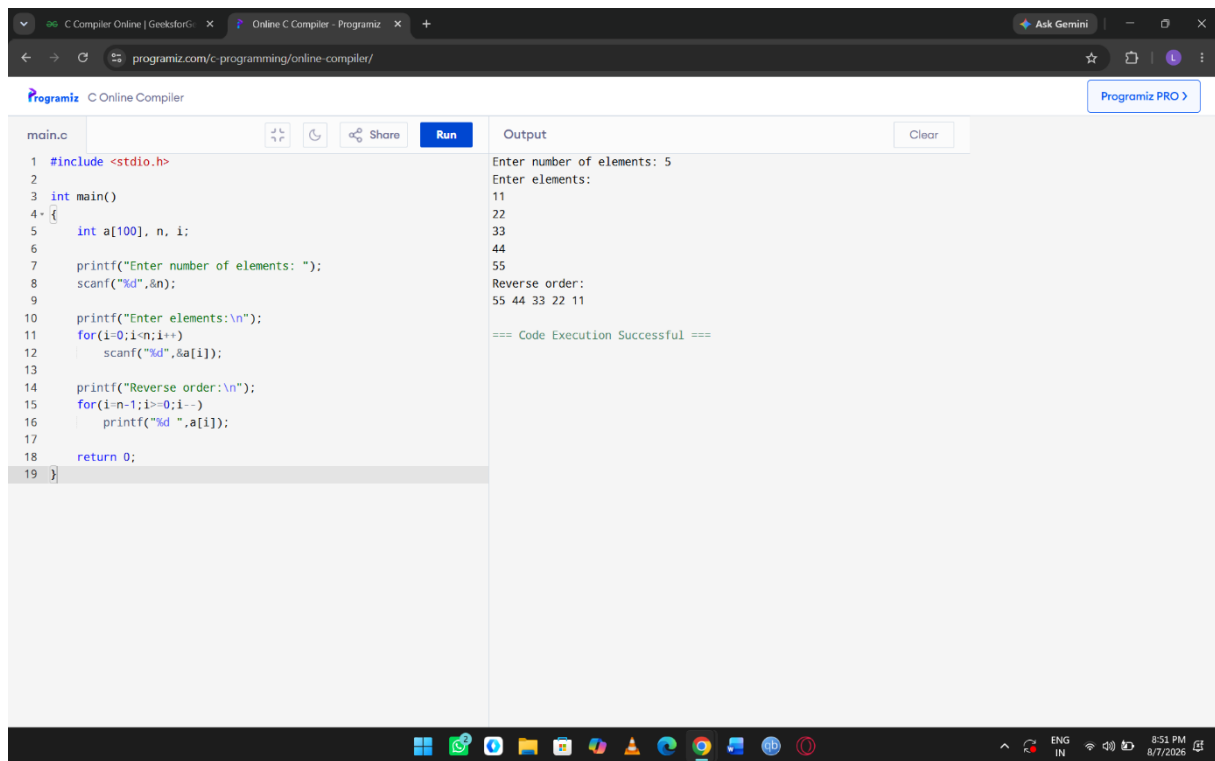
```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[10][10], b[10][10], c[10][10];
6     int i, j, r, col;
7
8     printf("Enter rows and columns: ");
9     scanf("%d%d", &r, &col);
10
11     printf("Enter first matrix:\n");
12     for(i=0; i<r; i++)
13         for(j=0; j<col; j++)
14             scanf("%d", &a[i][j]);
15
16     printf("Enter second matrix:\n");
17     for(i=0; i<r; i++)
18         for(j=0; j<col; j++)
19             scanf("%d", &b[i][j]);
20
21     printf("Sum of matrices:\n");
22     for(i=0; i<r; i++)
23     {
24         for(j=0; j<col; j++)
25         {
26             c[i][j] = a[i][j] + b[i][j];
27             printf("%d ", c[i][j]);
28         }
29         printf("\n");
30     }
```

Output:

```
Enter rows and columns: 2
2
Enter first matrix:
7
8
9
6
Enter second matrix:
1
2
3
4
Sum of matrices:
8 10
12 10

=== Code Execution Successful ===
```

2. Write a C program to read n number of values in an array and display them in reverse order.

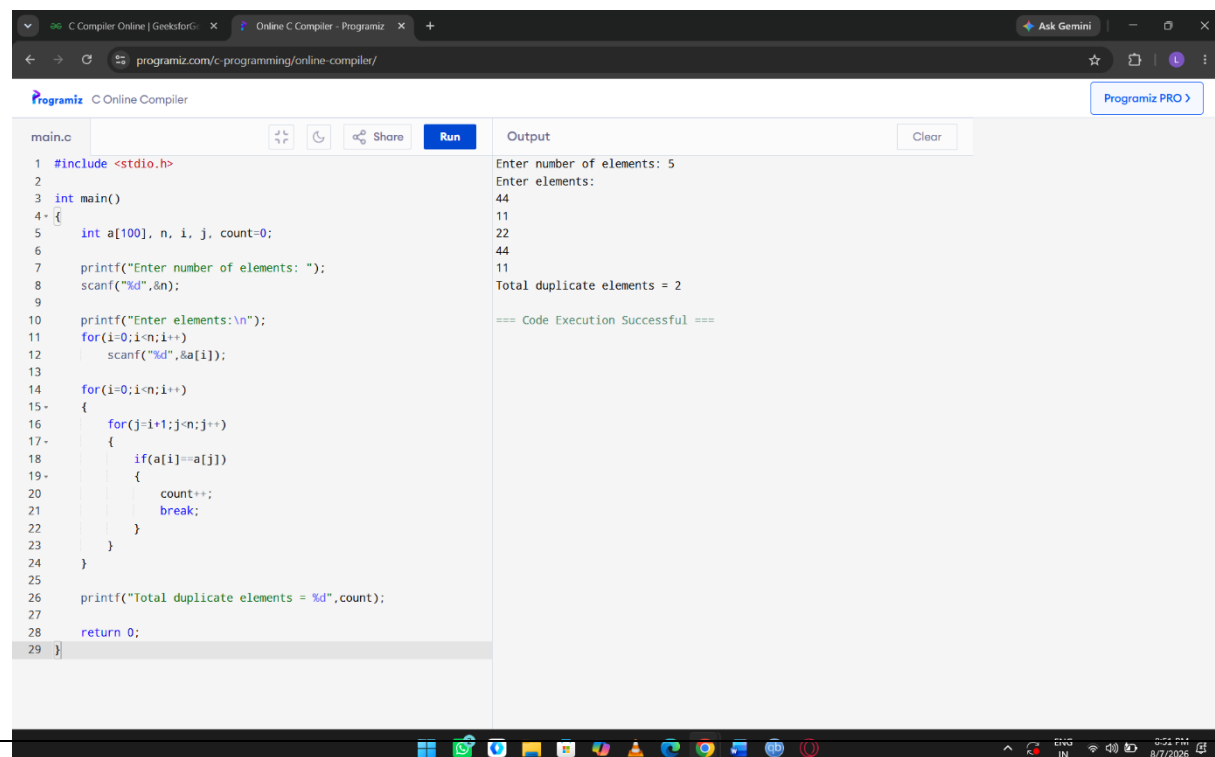


The screenshot shows a web browser with the URL `programiz.com/c-programming/online-compiler/`. The page displays the Programiz Online Compiler interface. On the left, the code for `main.c` is shown, and on the right, the output of the program is displayed.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, i;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    printf("Enter elements:\n");
11    for(i=0; i<n; i++)
12        scanf("%d", &a[i]);
13
14    printf("Reverse order:\n");
15    for(i=n-1; i>=0; i--)
16        printf("%d ", a[i]);
17
18    return 0;
19 }
```

The output shows the user entering 5 elements: 11, 22, 33, 44, 55. The program then displays the elements in reverse order: 55 44 33 22 11. The output also includes the text "=== Code Execution Successful ===".

3. Count the total number of duplicate elements in an array.

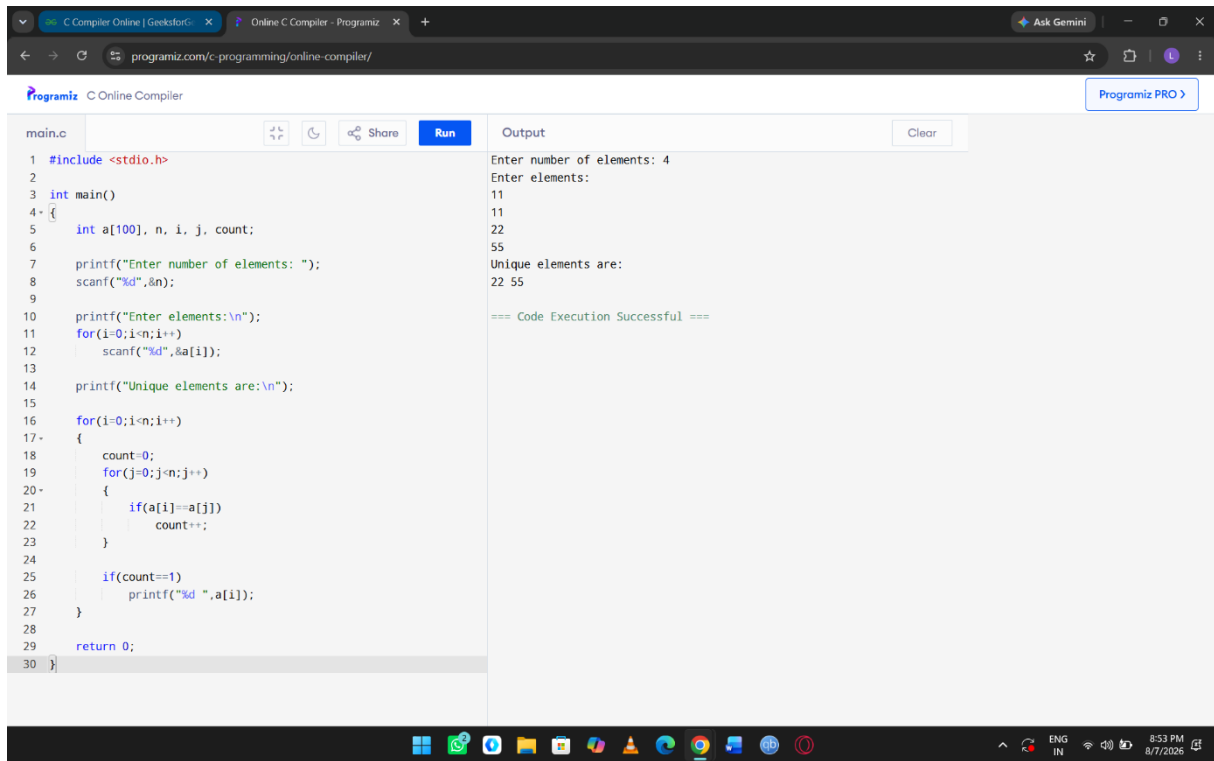


The screenshot shows the same online C compiler interface. The code for `main.c` is updated to count duplicate elements in an array. The output shows the user entering 5 elements: 44, 11, 22, 44, 11. The program then displays the total number of duplicate elements: 2. The output also includes the text "=== Code Execution Successful ===".

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, i, j, count=0;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    printf("Enter elements:\n");
11    for(i=0; i<n; i++)
12        scanf("%d", &a[i]);
13
14    for(i=0; i<n; i++)
15    {
16        for(j=i+1; j<n; j++)
17        {
18            if(a[i]==a[j])
19            {
20                count++;
21                break;
22            }
23        }
24    }
25
26    printf("Total duplicate elements = %d", count);
27
28    return 0;
29 }
```

The output shows the user entering 5 elements: 44, 11, 22, 44, 11. The program then displays the total number of duplicate elements: 2. The output also includes the text "=== Code Execution Successful ===".

4. Print unique elements in an array:



The screenshot shows the Programiz C Online Compiler interface. The code in `main.c` is as follows:

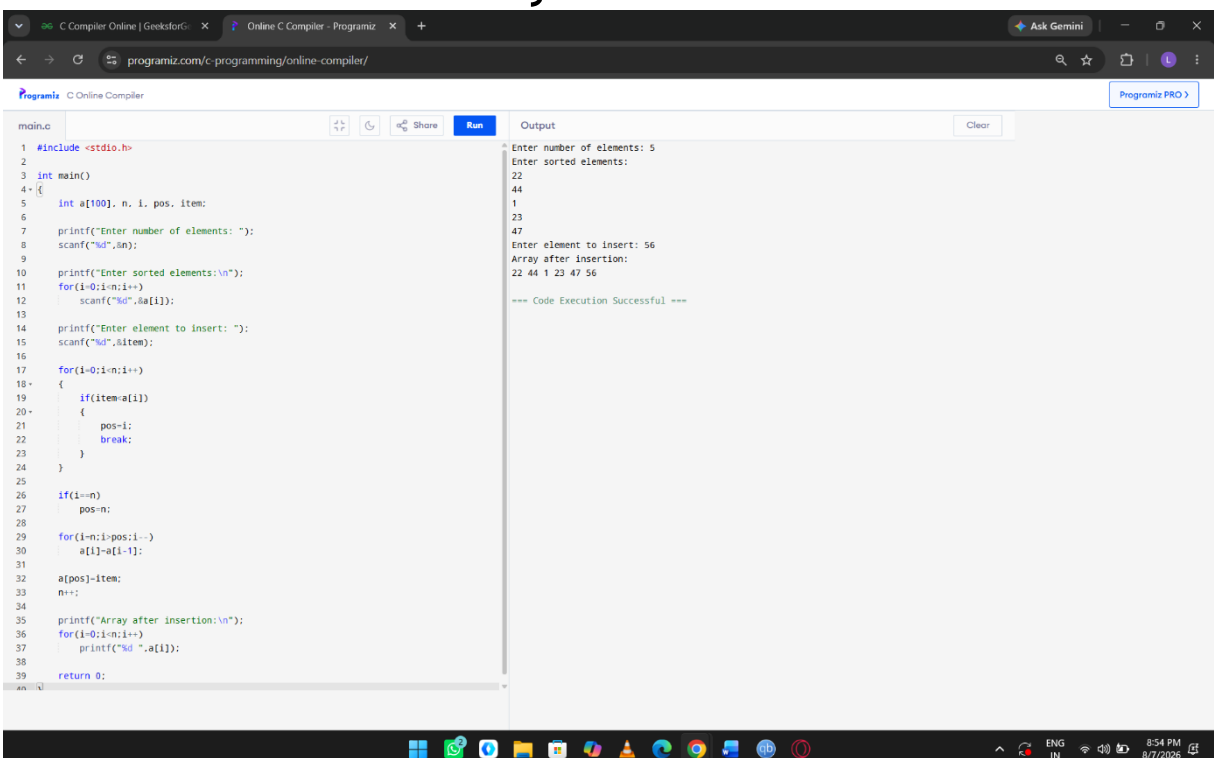
```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, i, j, count;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    printf("Enter elements:\n");
11    for(i=0; i<n; i++)
12        scanf("%d", &a[i]);
13
14    printf("Unique elements are:\n");
15
16    for(i=0; i<n; i++)
17    {
18        count=0;
19        for(j=0; j<n; j++)
20        {
21            if(a[i]==a[j])
22                count++;
23        }
24
25        if(count==1)
26            printf("%d ", a[i]);
27    }
28
29    return 0;
30 }
```

The output of the program is:

```
Enter number of elements: 4
Enter elements:
11
11
22
55
Unique elements are:
22 55

=== Code Execution Successful ===
```

5. Insert in sorted array



The screenshot shows the Programiz C Online Compiler interface. The code in `main.c` is as follows:

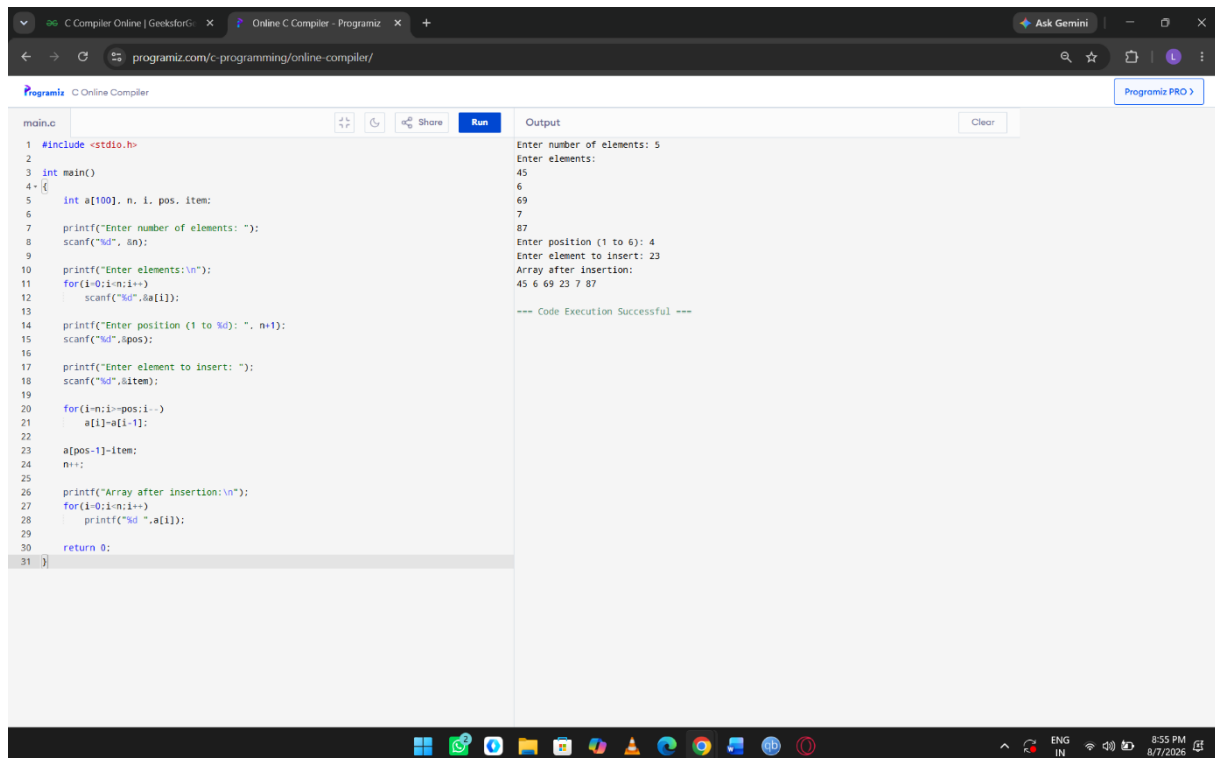
```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, i, pos, item;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    printf("Enter sorted elements:\n");
11    for(i=0; i<n; i++)
12        scanf("%d", &a[i]);
13
14    printf("Enter element to insert: ");
15    scanf("%d", &item);
16
17    for(i=0; i<n; i++)
18    {
19        if(item>a[i])
20        {
21            pos=i;
22            break;
23        }
24    }
25
26    if(i==n)
27        pos=n;
28
29    for(i=n-1; i>pos; i--)
30        a[i+1]=a[i];
31
32    a[pos]=item;
33    n++;
34
35    printf("Array after insertion:\n");
36    for(i=0; i<n; i++)
37        printf("%d ", a[i]);
38
39    return 0;
40 }
```

The output of the program is:

```
Enter number of elements: 5
Enter sorted elements:
22
44
1
23
47
Enter element to insert: 56
Array after insertion:
22 44 1 23 47 56

=== Code Execution Successful ===
```

6.Insert in sorted array.



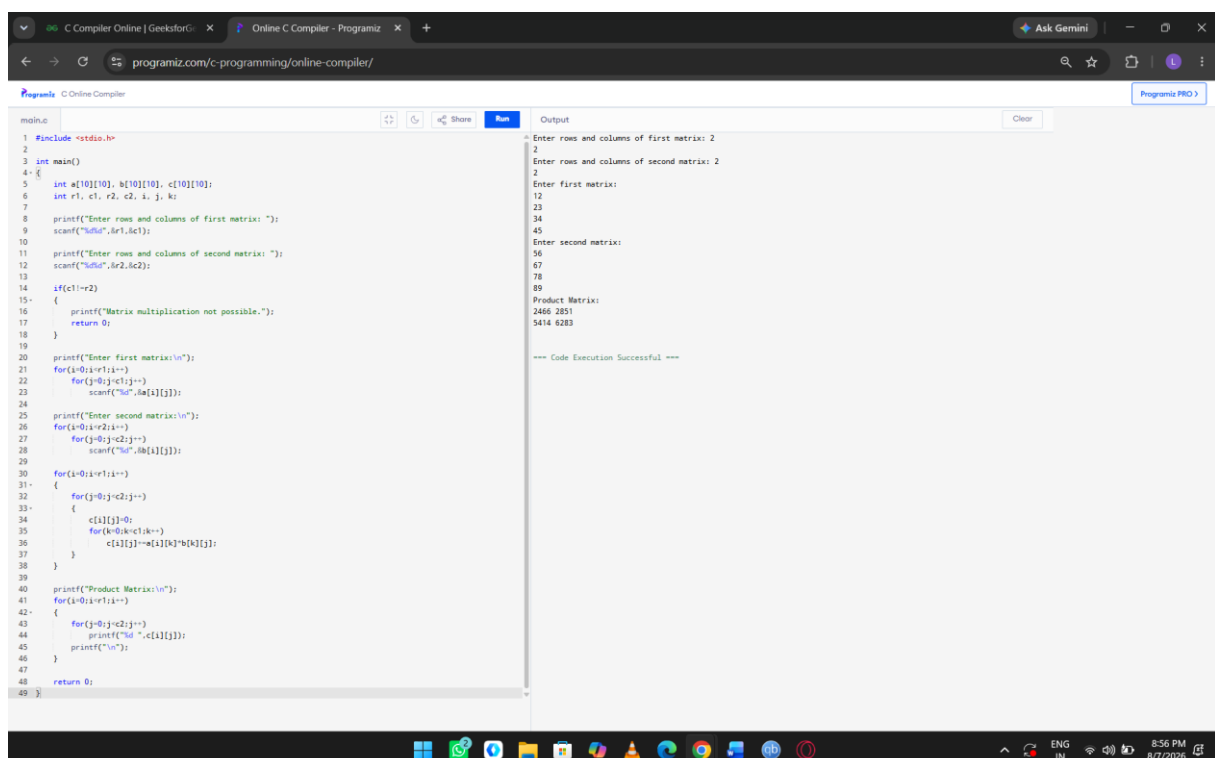
The screenshot shows a web browser with the URL `programiz.com/c-programming/online-compiler/`. The page displays a C program for inserting an element into a sorted array. The code is as follows:

```
main.c
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, i, pos, item;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    printf("Enter elements:\n");
11    for(i=0; i<n; i++)
12        scanf("%d", &a[i]);
13
14    printf("Enter position (1 to %d): ", n+1);
15    scanf("%d", &pos);
16
17    printf("Enter element to insert: ");
18    scanf("%d", &item);
19
20    for(i=n; i>=pos; i--)
21        a[i+1]=a[i];
22
23    a[pos]=item;
24    n++;
25
26    printf("Array after insertion:\n");
27    for(i=0; i<n; i++)
28        printf("%d ", a[i]);
29
30    return 0;
31 }
```

The output of the program is shown on the right:

```
Enter number of elements: 5
Enter elements:
45
69
7
87
Enter position (1 to 6): 4
Enter element to insert: 23
Array after insertion:
45 6 69 23 7 87
=== Code Execution Successful ===
```

7.Matrix multiplication.



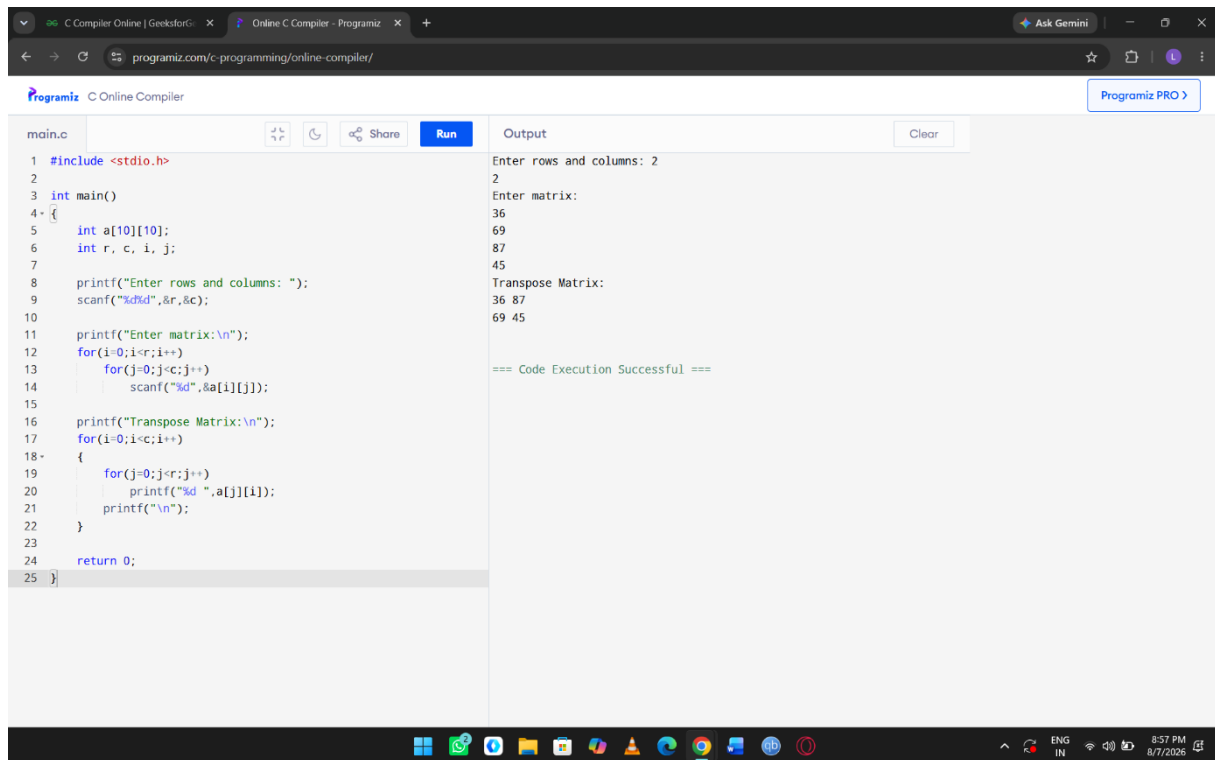
The screenshot shows a web browser with the URL `programiz.com/c-programming/online-compiler/`. The page displays a C program for matrix multiplication. The code is as follows:

```
main.c
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[10][10], b[10][10], c[10][10];
6     int r1, c1, r2, c2, i, j, k;
7
8     printf("Enter rows and columns of first matrix: ");
9     scanf("%d%d", &r1, &c1);
10
11    printf("Enter rows and columns of second matrix: ");
12    scanf("%d%d", &r2, &c2);
13
14    if(c1!=r2)
15    {
16        printf("Matrix multiplication not possible.");
17        return 0;
18    }
19
20    printf("Enter first matrix:\n");
21    for(i=0; i<r1; i++)
22        for(j=0; j<c1; j++)
23            scanf("%d", &a[i][j]);
24
25    printf("Enter second matrix:\n");
26    for(i=0; i<r2; i++)
27        for(j=0; j<c2; j++)
28            scanf("%d", &b[i][j]);
29
30    for(i=0; i<r1; i++)
31    {
32        for(j=0; j<c2; j++)
33        {
34            c[i][j]=0;
35            for(k=0; k<c1; k++)
36                c[i][j]+=a[i][k]*b[k][j];
37        }
38    }
39
40    printf("Product Matrix:\n");
41    for(i=0; i<r1; i++)
42    {
43        for(j=0; j<c2; j++)
44            printf("%d ", c[i][j]);
45        printf("\n");
46    }
47
48    return 0;
49 }
```

The output of the program is shown on the right:

```
Enter rows and columns of first matrix: 2
2
Enter rows and columns of second matrix: 2
2
Enter first matrix:
12
23
34
45
Enter second matrix:
56
67
78
89
Product Matrix:
2466 2851
5414 6283
=== Code Execution Successful ===
```

8. Matrix transpose.



The screenshot shows the Programiz C Online Compiler interface. The code in the editor is as follows:

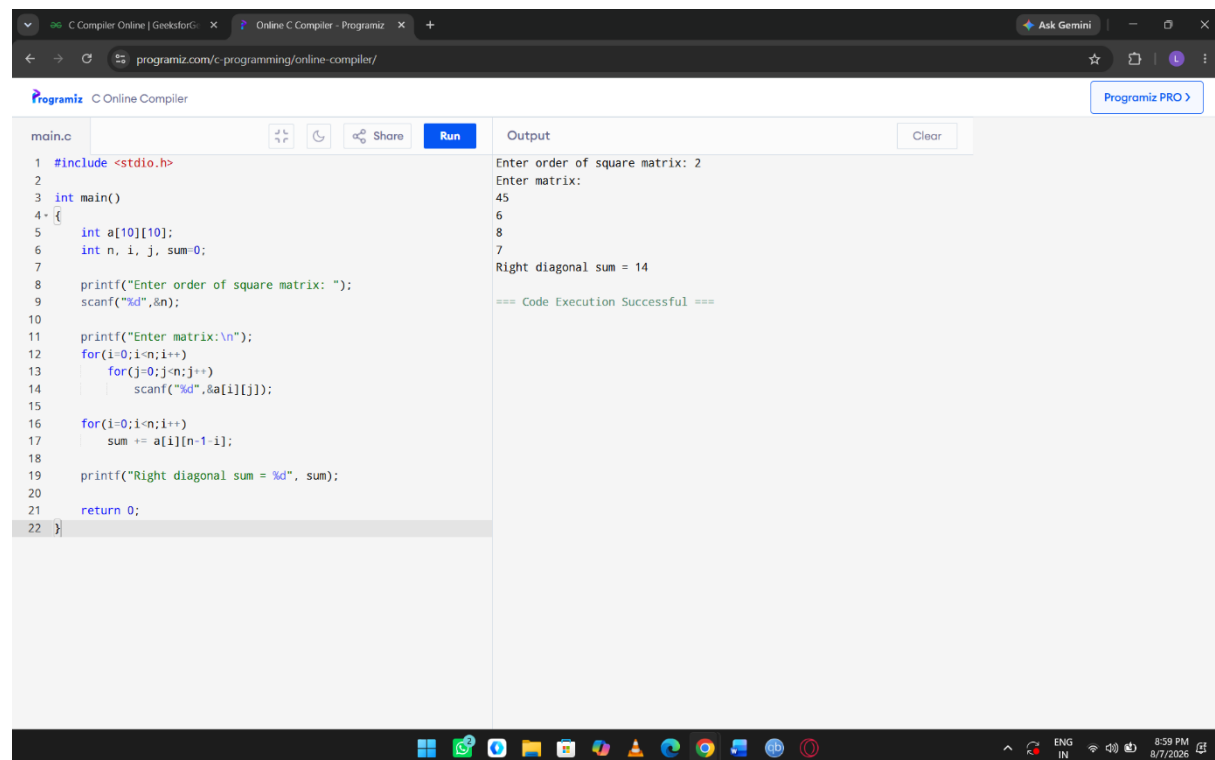
```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[10][10];
6     int r, c, i, j;
7
8     printf("Enter rows and columns: ");
9     scanf("%d%d", &r, &c);
10
11     printf("Enter matrix:\n");
12     for(i=0; i<r; i++)
13         for(j=0; j<c; j++)
14             scanf("%d", &a[i][j]);
15
16     printf("Transpose Matrix:\n");
17     for(i=0; i<c; i++)
18     {
19         for(j=0; j<r; j++)
20             printf("%d ", a[j][i]);
21         printf("\n");
22     }
23
24     return 0;
25 }
```

The output on the right shows the execution results:

```
Enter rows and columns: 2
2
Enter matrix:
36
69
87
45
Transpose Matrix:
36 87
69 45

=== Code Execution Successful ===
```

9. Right diagonal sum



The screenshot shows the Programiz C Online Compiler interface. The code in the editor is as follows:

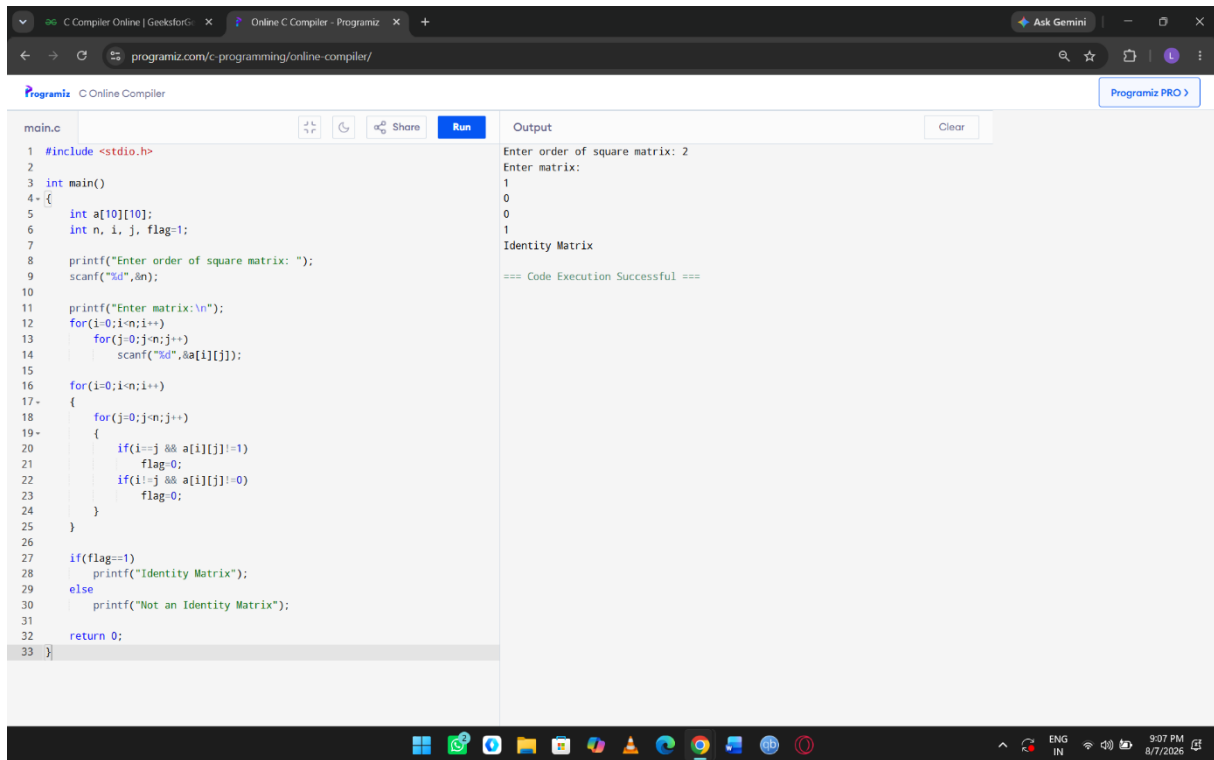
```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[10][10];
6     int n, i, j, sum=0;
7
8     printf("Enter order of square matrix: ");
9     scanf("%d", &n);
10
11     printf("Enter matrix:\n");
12     for(i=0; i<n; i++)
13         for(j=0; j<n; j++)
14             scanf("%d", &a[i][j]);
15
16     for(i=0; i<n; i++)
17         sum += a[i][n-1-i];
18
19     printf("Right diagonal sum = %d", sum);
20
21     return 0;
22 }
```

The output on the right shows the execution results:

```
Enter order of square matrix: 2
2
Enter matrix:
45
6
8
7
Right diagonal sum = 14

=== Code Execution Successful ===
```

10. Check for identity matrix



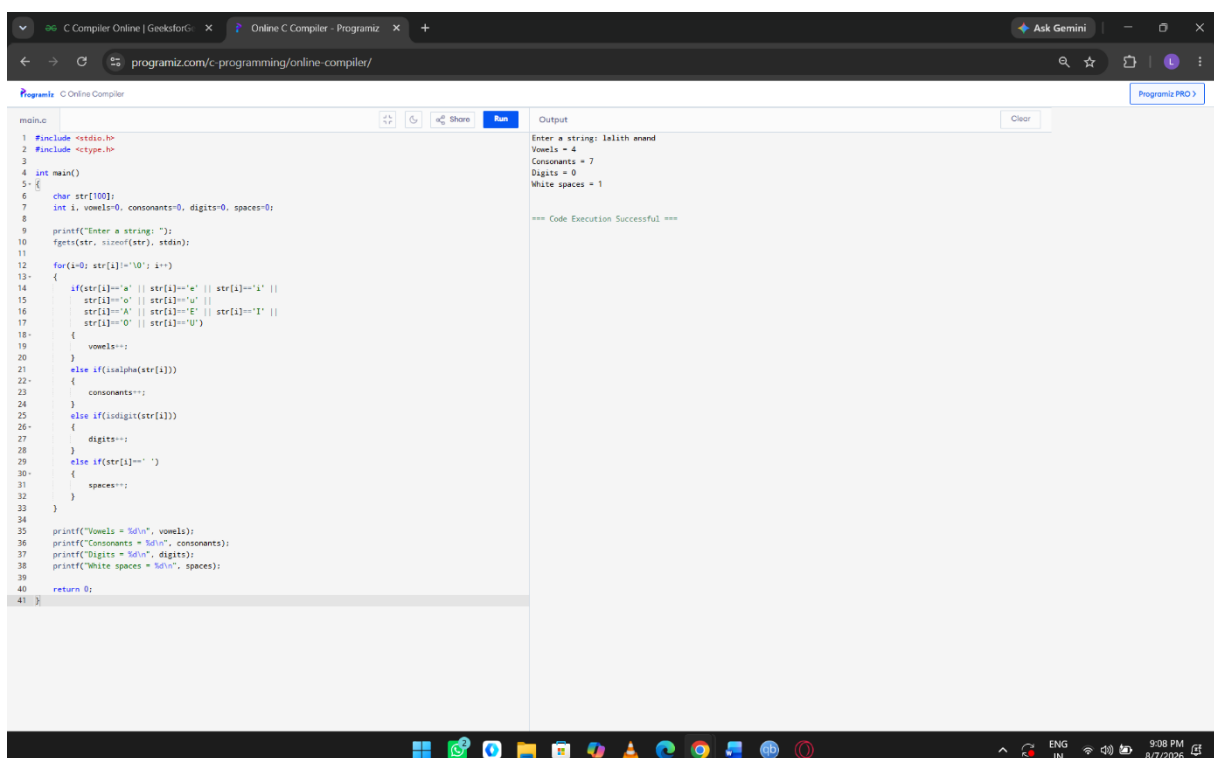
The screenshot shows a web browser with the URL `programiz.com/c-programming/online-compiler/`. The page displays a C program in the editor and its output in the right-hand pane. The C program is as follows:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[10][10];
6     int n, i, j, flag=1;
7
8     printf("Enter order of square matrix: ");
9     scanf("%d", &n);
10
11     printf("Enter matrix:\n");
12     for(i=0; i<n; i++)
13         for(j=0; j<n; j++)
14             scanf("%d", &a[i][j]);
15
16     for(i=0; i<n; i++)
17     {
18         for(j=0; j<n; j++)
19         {
20             if(i==j && a[i][j]!=1)
21                 flag=0;
22             if(i!=j && a[i][j]!=0)
23                 flag=0;
24         }
25     }
26
27     if(flag==1)
28         printf("Identity Matrix");
29     else
30         printf("Not an Identity Matrix");
31
32     return 0;
33 }
```

The output pane shows the following text:

```
Enter order of square matrix: 2
Enter matrix:
1
0
0
1
Identity Matrix
=== Code Execution Successful ===
```

11. To find out number of vowels, consonants, digits, and white spaces



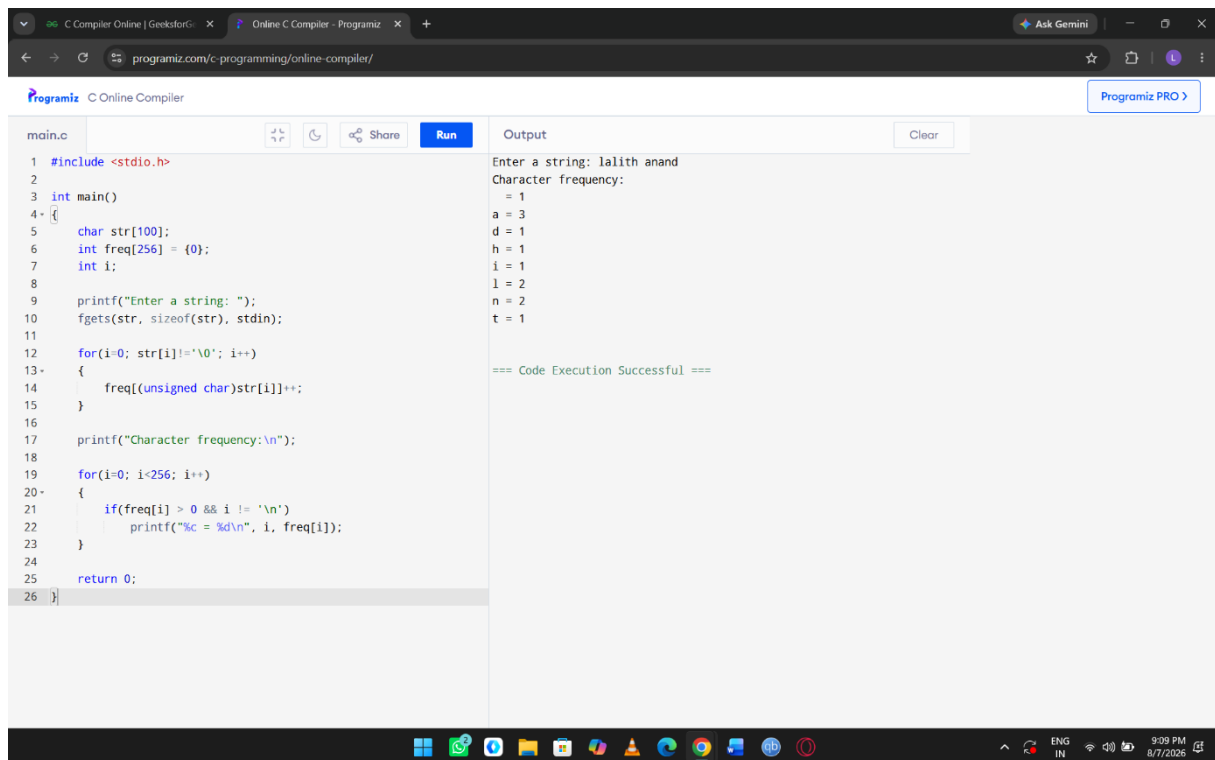
The screenshot shows a web browser with the URL `programiz.com/c-programming/online-compiler/`. The page displays a C program in the editor and its output in the right-hand pane. The C program is as follows:

```
1 #include <stdio.h>
2 #include <ctype.h>
3
4 int main()
5 {
6     char str[100];
7     int i, vowels=0, consonants=0, digits=0, spaces=0;
8
9     printf("Enter a string: ");
10    fgets(str, sizeof(str), stdin);
11
12    for(i=0; str[i]!='\0'; i++)
13    {
14        if(str[i]=='a' || str[i]=='e' || str[i]=='i' ||
15           str[i]=='o' || str[i]=='u' ||
16           str[i]=='A' || str[i]=='E' || str[i]=='I' ||
17           str[i]=='O' || str[i]=='U')
18            vowels++;
19        else if(isalpha(str[i]))
20            consonants++;
21        else if(isdigit(str[i]))
22            digits++;
23        else if(str[i]==' ' || str[i]=='\n')
24            spaces++;
25    }
26
27    printf("Vowels = %d\n", vowels);
28    printf("Consonants = %d\n", consonants);
29    printf("Digits = %d\n", digits);
30    printf("White spaces = %d\n", spaces);
31
32    return 0;
33 }
```

The output pane shows the following text:

```
Enter a string: lalith anand
Vowels = 4
Consonants = 7
Digits = 0
White spaces = 1
=== Code Execution Successful ===
```

12. Frequency of character in a string.



The screenshot shows the Programiz C Online Compiler interface. The code in the editor is as follows:

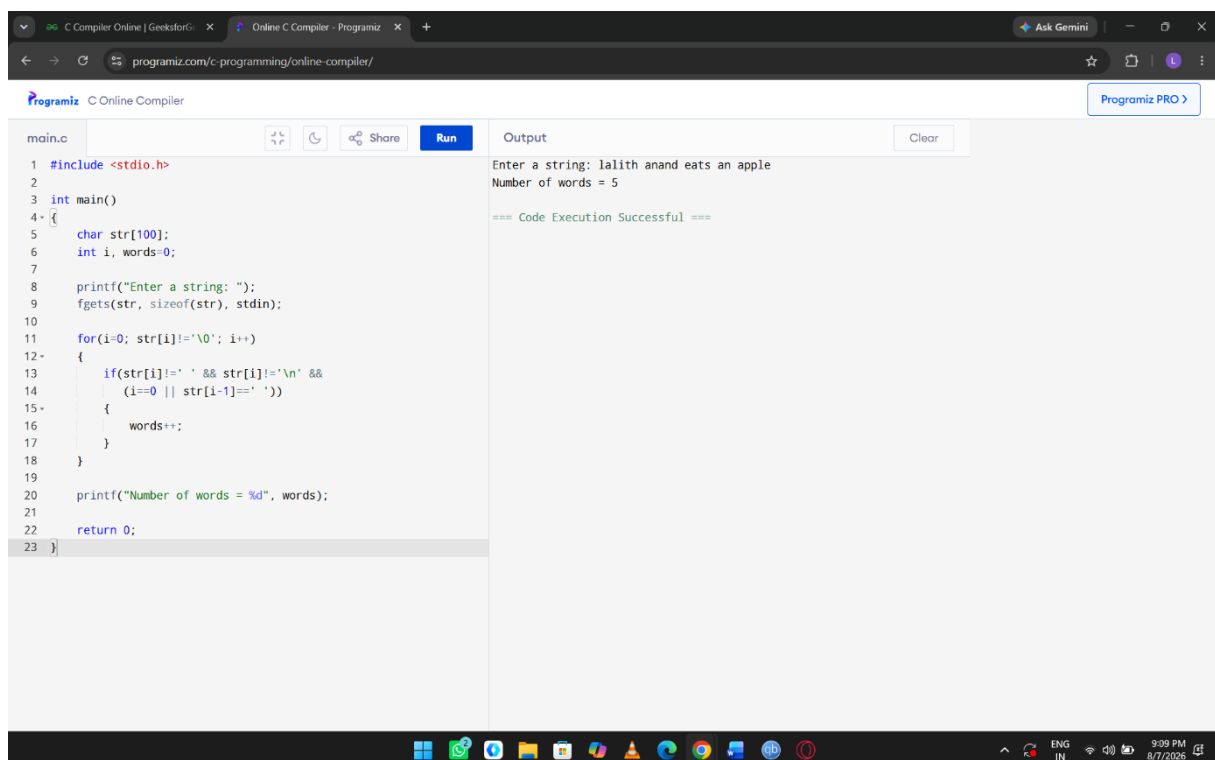
```
1 #include <stdio.h>
2
3 int main()
4 {
5     char str[100];
6     int freq[256] = {0};
7     int i;
8
9     printf("Enter a string: ");
10    fgets(str, sizeof(str), stdin);
11
12    for(i=0; str[i]!='\0'; i++)
13    {
14        freq[(unsigned char)str[i]]++;
15    }
16
17    printf("Character frequency:\n");
18
19    for(i=0; i<256; i++)
20    {
21        if(freq[i] > 0 && i != '\n')
22            printf("%c = %d\n", i, freq[i]);
23    }
24
25    return 0;
26 }
```

The output on the right shows the execution results for the input string "lalith anand":

```
Enter a string: lalith anand
Character frequency:
= 1
a = 3
d = 1
h = 1
i = 1
l = 2
n = 2
t = 1

=== Code Execution Successful ===
```

13. Count words in a string.



The screenshot shows the Programiz C Online Compiler interface. The code in the editor is as follows:

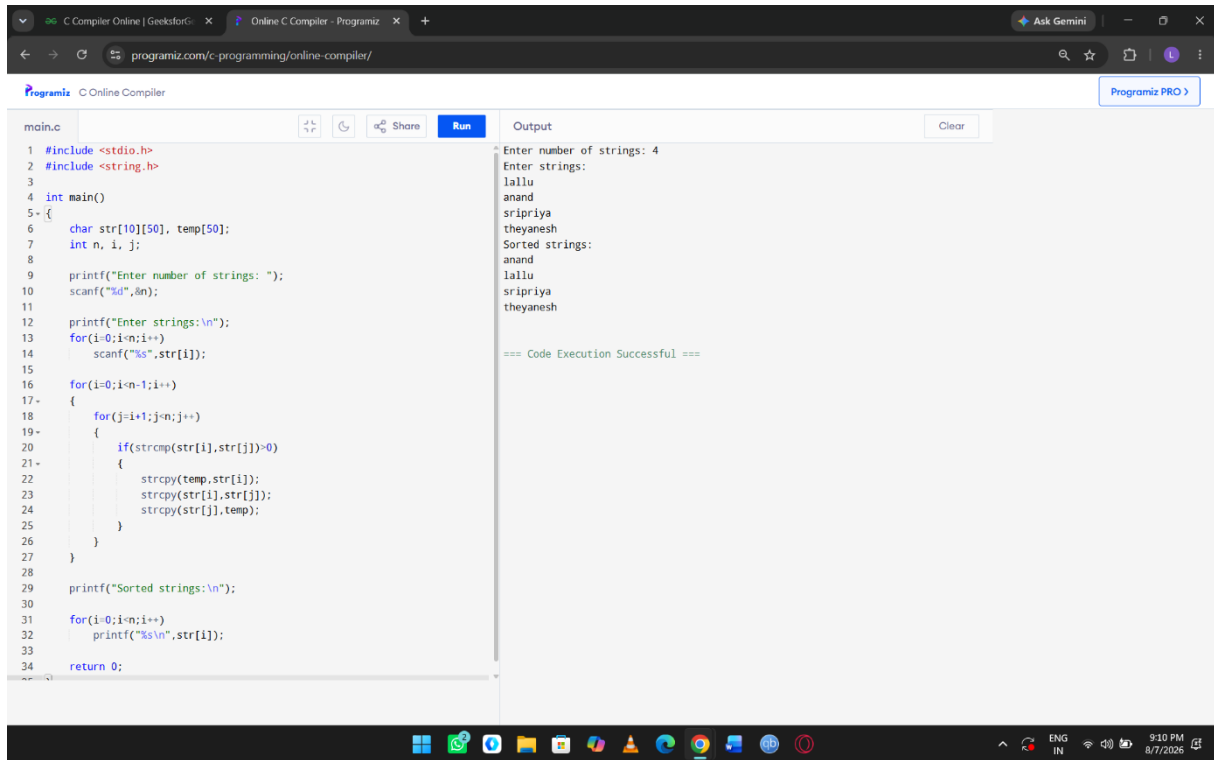
```
1 #include <stdio.h>
2
3 int main()
4 {
5     char str[100];
6     int i, words=0;
7
8     printf("Enter a string: ");
9     fgets(str, sizeof(str), stdin);
10
11    for(i=0; str[i]!='\0'; i++)
12    {
13        if(str[i]!=' ' && str[i]!='\n' &&
14           (i==0 || str[i-1]!=' '))
15        {
16            words++;
17        }
18    }
19
20    printf("Number of words = %d", words);
21
22    return 0;
23 }
```

The output on the right shows the execution results for the input string "lalith anand eats an apple":

```
Enter a string: lalith anand eats an apple
Number of words = 5

=== Code Execution Successful ===
```

14.Sort string array



The screenshot shows a web browser window with the URL `programiz.com/c-programming/online-compiler/`. The page displays a C program for sorting a string array using bubble sort. The code is as follows:

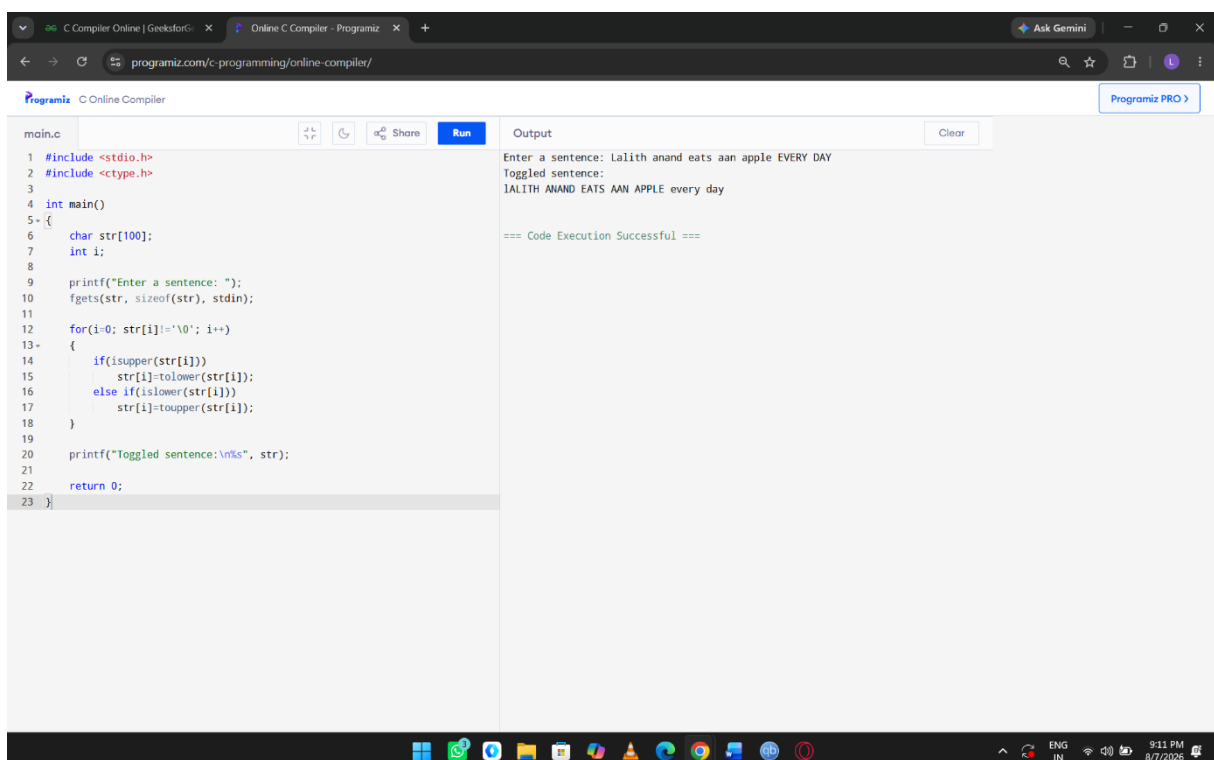
```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char str[10][50], temp[50];
7     int n, i, j;
8
9     printf("Enter number of strings: ");
10    scanf("%d", &n);
11
12    printf("Enter strings:\n");
13    for(i=0; i<n; i++)
14        scanf("%s", str[i]);
15
16    for(i=0; i<n-1; i++)
17    {
18        for(j=i+1; j<n; j++)
19        {
20            if(strcmp(str[i], str[j])>0)
21            {
22                strcpy(temp, str[i]);
23                strcpy(str[i], str[j]);
24                strcpy(str[j], temp);
25            }
26        }
27    }
28
29    printf("Sorted strings:\n");
30    for(i=0; i<n; i++)
31        printf("%s\n", str[i]);
32
33    return 0;
34 }
```

The output of the program is shown on the right side of the compiler interface:

```
Enter number of strings: 4
Enter strings:
lallu
anand
sripriya
theyanesh
Sorted strings:
anand
lallu
sripriya
theyanesh

=== Code Execution Successful ===
```

15.Toggle case of a sentence.



The screenshot shows a web browser window with the URL `programiz.com/c-programming/online-compiler/`. The page displays a C program for toggling the case of a sentence. The code is as follows:

```
1 #include <stdio.h>
2 #include <ctype.h>
3
4 int main()
5 {
6     char str[100];
7     int i;
8
9     printf("Enter a sentence: ");
10    fgets(str, sizeof(str), stdin);
11
12    for(i=0; str[i]!='\0'; i++)
13    {
14        if(isupper(str[i]))
15            str[i]=tolower(str[i]);
16        else if(islower(str[i]))
17            str[i]=toupper(str[i]);
18    }
19
20    printf("Toggled sentence:\n%s", str);
21
22    return 0;
23 }
```

The output of the program is shown on the right side of the compiler interface:

```
Enter a sentence: Lalith anand eats aan apple EVERY DAY
Toggled sentence:
IALITH ANAND EATS AAN APPLE every day

=== Code Execution Successful ===
```